Quantification: I534_z19c1-2
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PRE-VALIDATION: no number here has been checked against manual (ImageJ) counts.
PIXEL UNITS: lengths px, areas px^2, densities px^-1. The images carry no micron calibration.
Statistics are over interior lacunae only (those not touching the frame edge).

Per image
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measure	value
lacuna count	17
interior lacuna count	17
frame-edge lacuna count	0

Interior lacunae only: mean, median and sample SD (n = 17)
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measure	kind	mean	median	sd	units
area_px2	lacuna shape	2780.41	2632	1567.89	px^2
major_axis_length_px	lacuna shape	87.7017	86.6013	28.4991	px
minor_axis_length_px	lacuna shape	45.382	44.321	11.9863	px
aspect_ratio	lacuna shape	1.9779	2.1186	0.5509	unitless
eccentricity	lacuna shape	0.8044	0.8816	0.1488	unitless
solidity	lacuna shape	0.7514	0.7614	0.1011	unitless
roots_count	headline	11	9	4.3875	count
ring_length_r30_px	headline	277.412	271	80.3967	px
ring_length_r60_px	reported	566.353	608	159.955	px
owned_length_px	ownership-dependent	756.187	462.676	793.657	px
edge_count	ownership-dependent	36.2941	20	38.1047	count
mean_edge_length_px	ownership-dependent	20.8502	20.1545	3.4492	px
perimeter_px	normalised	296.072	283.463	92.9543	px
ring_area_r30_px2	normalised	10022.6	9930	2233.52	px^2
ring_area_r60_px2	normalised	21765.1	21920	3653.35	px^2
in_frame_fraction_r30	normalised	1	1	0	unitless
in_frame_fraction_r60	normalised	0.9933	1	0.0271	unitless
ring_density_r30	normalised	0.0282623	0.0257575	0.00929416	px^-1
ring_density_r60	normalised	0.0261903	0.0253068	0.00737294	px^-1
roots_per_100px_perimeter	normalised	3.9301	3.2352	1.7641	per 100 px
ring_attached_length_r30_px	network v2	248.647	248	69.4235	px
ring_attached_length_r60_px	network v2	386.059	405	134.975	px
ring_length_w_r30_px	network v2	314.868	305.35	90.7398	px
ring_length_w_r60_px	network v2	642.381	691.512	181.016	px
sholl_crossings_r10	network v2	6.2941	6	1.7235	count
sholl_crossings_r20	network v2	7.2353	7	3.2506	count
sholl_crossings_r30	network v2	7	7	2.6458	count
loop_count	loops, report only	2.2941	0	5.4973	count
loop_edge_share	loops, report only	0.0965	0	0.1479	share
ring_width_mean_r30_px	width	7.5241	7.4688	0.5741	px

Per field
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measure	value	units
analysed_area_px2	1.00131e+06	px^2
total_skeleton_length_px	22771	px
canalicular_length_density_per_px	0.0227412	px^-1
junction_count	440	count
junction_density_per_px2	0.00043942	px^-2
lacuna_count	17	count

lacunae_per_px2	1.698e-05	px^-2
skeleton_component_count	349	count
mean_component_length_px	65.2464	px
median_component_length_px	23	px
field_length_density_w_per_px	0.0258057	px^-1
field_density_without_flagged_per_px	0.02445	px^-1
field_density_in_roi_per_px		px^-1
unattached_length_px	429	px
unattached_component_count	25	count
canalicular_length_density_attached_per_px	0.0227412	px^-1
field_length_density_with_unattached_per_px	0.0231697	px^-1
loop_count	46	count
loop_edge_share	0.135	share
edge_count	1207	count

Canalicular width, the new measure

measure	value	units
width_mean_px	8.057	px
width_median_px	8	px
width_p10_px	4	px
width_p90_px	11.6619	px
width_pixels_used	21786	count

Canalicular width: 2 x the Euclidean distance transform of the canalicular mask at each skeleton pixel. Measured only on skeleton pixels on the canalicular mask: the lacuna buffer and the dilated vascular mask are excluded. The threads are about 2 to 6 px wide, so the width is close to the pixel grid and quantised in steps of about 1 px, it moves with the threshold (a lower cut widens every thread), and it has not been validated against any manual measurement.

Lacuna shape

lacuna_id	area_px2	major_axis_length_px	minor_axis_length_px	aspect_ratio	eccentricity	solidity	orientation_rad	centroid_row_px	centroid_col_px	on_border
1	1823	56.4889	49.8735	1.1326	0.4696	0.8271	-1.1579	125.003	757.529	False
2	1637	57.8693	48.4179	1.1952	0.5477	0.7276	-0.0784	295.51	896.101	False
3	1291	47.4215	38.1837	1.2419	0.593	0.8741	0.4227	324.111	336.189	False
4	6026	129.945	61.3358	2.1186	0.8816	0.8485	-1.1185	377.349	511.866	False
5	2978	75.2858	55.7926	1.3494	0.6714	0.7831	-0.8753	453.94	606.7	False
6	3737	84.848	59.7745	1.4195	0.7097	0.8338	-0.4377	493.364	441.851	False
7	820	51.4532	24.5895	2.0925	0.8784	0.7614	1.0688	505.329	211.1	False
8	3553	104.824	44.321	2.3651	0.9062	0.8721	0.2384	581.822	856.943	False
9	4782	122.937	62.4014	1.9701	0.8616	0.6511	1.4416	592.132	630.596	False
10	1615	90.9802	31.7451	2.866	0.9372	0.5886	-0.9716	606.27	724.946	False
11	2632	101.535	38.4077	2.6436	0.9257	0.6745	-1.3449	587.391	245.703	False
12	5166	140.866	61.0892	2.3059	0.9011	0.6428	0.013	686.162	69.5337	False
13	1712	87.747	36.8261	2.3827	0.9077	0.6786	-0.5403	724.265	671.19	False
14	914	54.9878	33.7201	1.6307	0.7899	0.5745	-0.9811	788.83	430.015	False
15	2692	86.6013	40.7838	2.1234	0.8822	0.8628	-1.1084	820.32	99.4658	False
16	1680	84.8432	32.3344	2.6239	0.9245	0.7391	-0.8417	827.977	860.805	False
17	4209	112.297	51.8973	2.1638	0.8868	0.8336	-0.6033	855.455	275.629	False

Network per lacuna

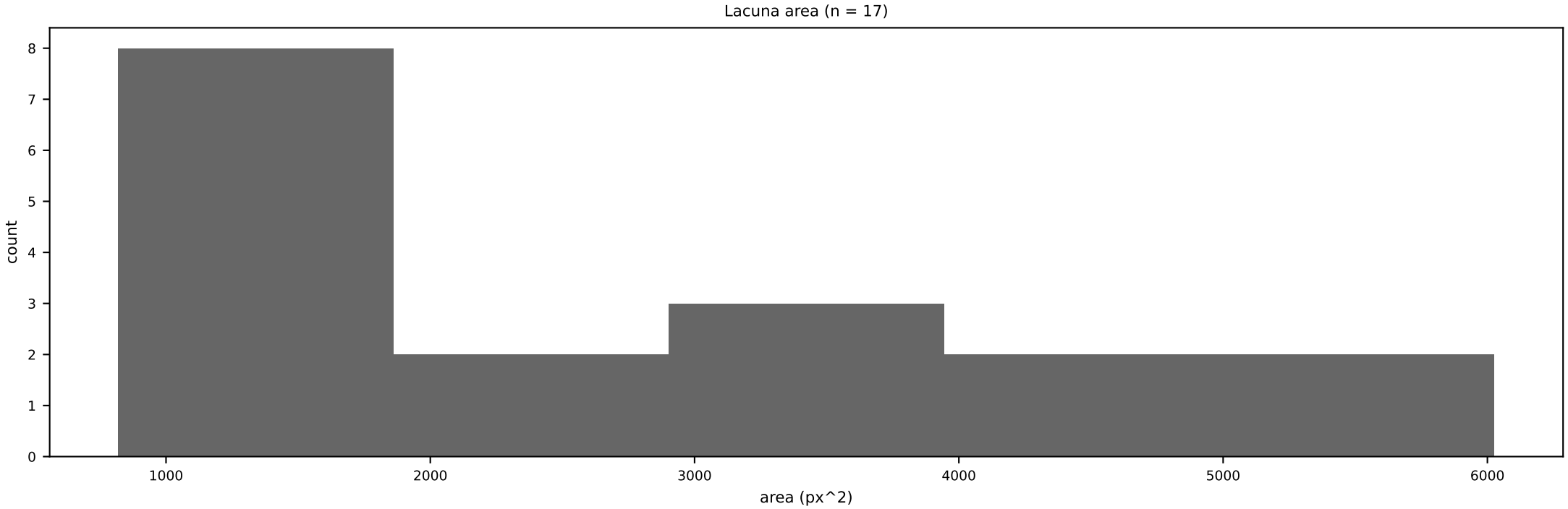
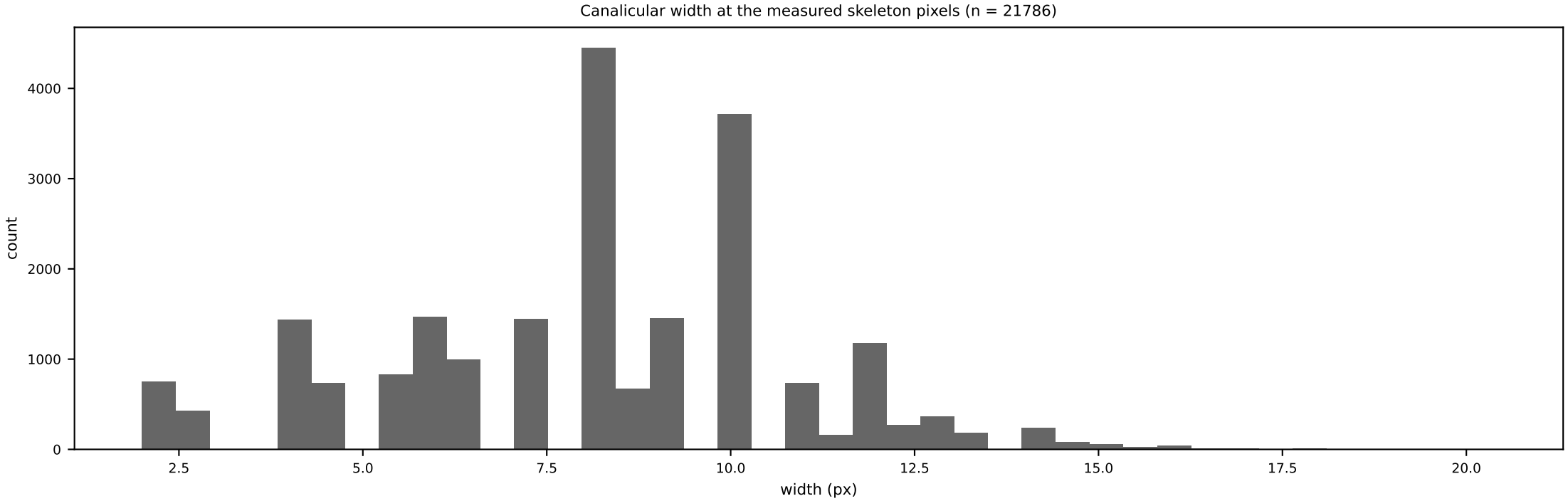
lacuna_id	roots_count	ring_length_r30_px	ring_length_r60_px	owned_length_px	edge_count	mean_edge_length_px	ring_width_mean_r30_px
1	4	143	525	341.593	17	20.0937	7.2097
2	7	136	333	403.09	20	20.1545	6.6193

3	9	269	718	2573.1	121	21.2653	6.7737
4	20	382	645	554.931	36	15.4148	7.0925
5	9	239	336	357.635	20	17.8817	8.132
6	15	237	506	307.764	17	18.1038	7.4688
7	13	378	666	704.37	45	15.6527	8.31
8	15	328	888	2984.05	146	20.4387	8.1696
9	14	312	526	374.635	19	19.7176	7.9321
10	7	176	268	120.125	7	17.1607	7.936
11	17	368	607	475.86	21	22.66	7.4204
12	14	397	639	447.274	18	24.8486	7.9251
13	8	322	608	915.293	35	26.1512	7.1883
14	12	249	609	952.377	38	25.0625	7.5833
15	8	236	400	298.392	15	19.8928	6.9221
16	6	271	659	462.676	18	25.7042	8.3868
17	9	273	695	582.014	24	24.2506	6.8395

Normalised per lacuna

lacuna_id	perimeter_px	ring_area_r30_px2	ring_area_r60_px2	in_frame_fraction_r30	in_frame_fraction_r60	ring_density_r30	ring_density_r60	roots_per_100px_perimeter
1	272.936	8396	22205	1	1	0.0170319	0.0236433	1.4655
2	283.463	8668	22573	1	1	0.0156899	0.0147521	2.4695
3	164.51	7220	19561	1	1	0.0372576	0.0367057	5.4708
4	382.854	12834	23568	1	1	0.0297647	0.0273676	5.2239
5	274.735	10367	18963	1	1	0.0230539	0.0177187	3.2759
6	300.049	10385	21896	1	1	0.0228214	0.0231092	4.9992
7	155.717	6688	15181	1	1	0.0565191	0.0438706	8.3485
8	288.735	10625	24530	1	1	0.0308706	0.0362006	5.1951
9	439.83	12113	21276	1	1	0.0257575	0.0247227	3.183
10	289.848	8213	14035	1	1	0.0214294	0.0190951	2.4151
11	316.777	10912	23859	1	1	0.0337243	0.0254411	5.3666
12	533.263	15675	28249	1	0.8882	0.025327	0.0226203	2.6253
13	263.664	9930	21920	1	1	0.032427	0.0277372	3.0342
14	215.723	8071	20618	1	1	0.0308512	0.0295373	5.5627
15	247.279	9530	20421	1	0.9977	0.0247639	0.0195877	3.2352
16	251.22	9124	23689	1	1	0.0297019	0.0278188	2.3883
17	352.617	11633	27463	1	1	0.0234677	0.0253068	2.5523

Network v2 per lacuna

lacuna_id	ring_attached_length_r30_px	ring_attached_length_r60_px	ring_length_w_r30_px	ring_length_w_r60_px	sholl_crossings_r10	sholl_crossings_r20	sholl_crossings_r30
1	126	205	163.853	608.299	4	7	7
2	108	149	151.024	366.149	4	1	4
3	232	469	305.35	816.796	4	9	9
4	326	486	428.433	725.642	8	12	10
5	230	286	276.764	382.191	5	6	3
6	222	282	265.794	568.973	6	5	4
7	341	469	423.563	739.144	8	12	7
8	304	680	365.007	997.666	7	9	13
9	286	345	346.664	597.6	6	3	5
10	176	256	199.995	306.735	4	3	4
11	317	405	418.647	691.512	8	10	7
12	338	448	453.132	730.068	9	11	10
13	287	439	376.676	700.269	8	9	8
14	249	486	288.836	691.127	5	4	5
15	199	250	270.007	457.517	6	8	7
16	248	518	298.309	729.6	8	7	8
17	238	390	320.706	811.193	7	7	8

I534_z19c1-2: distributions, pre-validation, px